# Design Review DR#1: Enterprise Knowledge Assistant (EKA)

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## What I am building

An Enterprise Knowledge Assistant that answers employee questions about HR, IT and operational policy from approved internal documents. Every answer carries citations, a confidence score and its cost. When evidence is missing or confidence falls below 0.40, the assistant escalates to the HR Service Desk or IT Helpdesk instead of guessing. Milestone 1 locks the scope, corpus, evaluation set and API contract; retrieval and generation follow. Scope is single-turn English question answering; transactions, personal records and legal advice are out.

## Who it is for

New joiners who need quick answers; managers who need cross-document guidance they can forward with a source; HR Business Partners and the IT Service Desk, who want fewer repeat tickets and cleaner hand-offs; and the executive sponsor, who needs measurable outcomes with controlled risk. Five named personas are in the stakeholder map.

## Current architecture

In place: a FastAPI service with bearer-token authentication, structured request logging and cost tracking; a frozen Answer schema v1 (content, cost, retries, confidence, sources, version); a placeholder handler that escalates every question; a 16-document synthetic corpus (HR 5, IT 6, Operations 5); a 20-question golden set; and an offline baseline harness. Next: hybrid retrieval, reranking, grounded generation and an answer-support check.

## KPI targets

| KPI | Baseline | Target |
|---|---|---|
| **K1** Question resolution rate | 38% (estimate) | 70% within 6 months |
| **K2** Answer accuracy | 25% (measured, lexical baseline) | 90% of answerable questions; every hard question escalated correctly |
| **K3** Knowledge tickets per month | 9,000 (estimate) | 30% fewer within 6 months |

## Key trade-offs

**Accuracy over coverage:** abstaining lowers resolution early on but avoids confident errors. **Quality versus latency and cost:** reranking and support checks add seconds and spend; targets are p95 under 8 seconds and under USD 0.03 per answer. **Build versus buy:** build the pipeline and evaluation, consume the model as a service. **Freshness versus stability:** re-indexing on publication keeps answers current but briefly lags a policy change. **Contract stability:** a five-field schema now, additive fields later.

## Uncertainties and risks

The K1 and K3 baselines are estimates that need real ticket data. The baseline passes only 25% of the golden set and none of the cross-document questions, so the gap to 90% is large and unproven. Twenty questions is a small sample. The main risks are stale or conflicting documents and leakage of Restricted data; mitigations are content owners, version metadata, Internal-only indexing and monthly expert audits. Open questions: which approved model and region, and who signs off escalation wording.

Estimates are illustrative planning assumptions. The accuracy baseline comes from scripts/run_rag_eval.py --fake.